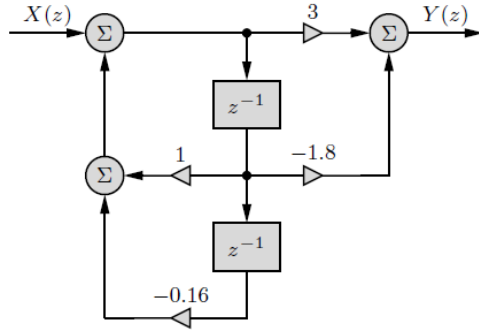
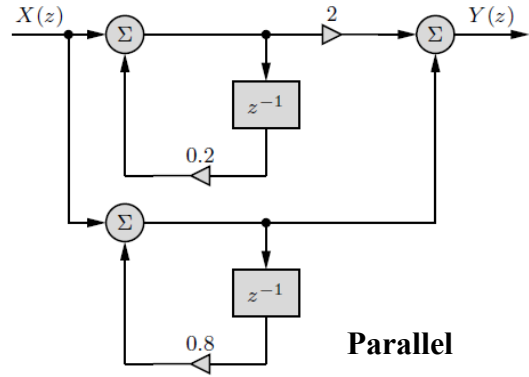


Answer Keys to HW 10

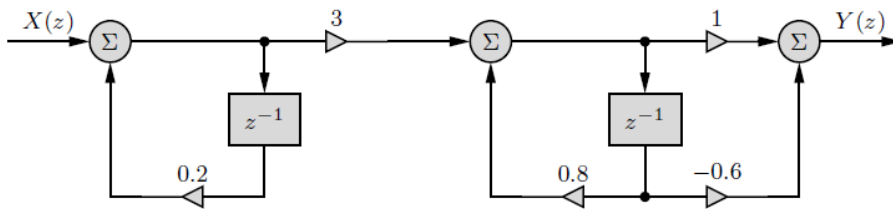
1)



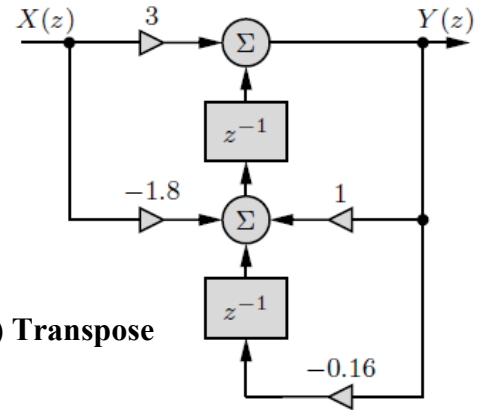
Direct Form II



Parallel



Cascade



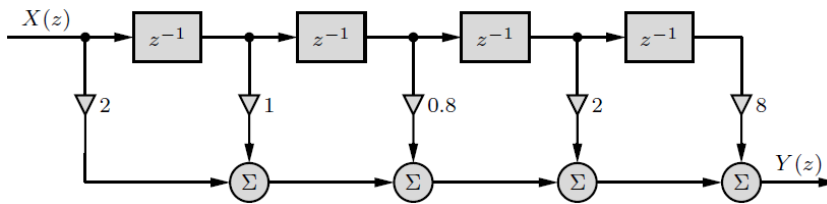
b) Transpose

2) a) $y_{\text{zsr,a}}[n] = [0.4856(e)^n - 0.4170(0.8)^n - 0.0685(-0.2)^n] u[n]$

b) $y_{\text{zsr,b}}[n] = \frac{25}{33}(2)^n u[-n-1] + \left[\frac{2}{3}(0.8)^n + \frac{1}{11}(-0.2)^n \right] u[n]$

c) $y_{\text{zsr,c}}[n] = \frac{25}{33}(2)^n u[-n-1] + [0.4856(e)^n + 0.2496(0.8)^n + 0.0224(-0.2)^n] u[n]$

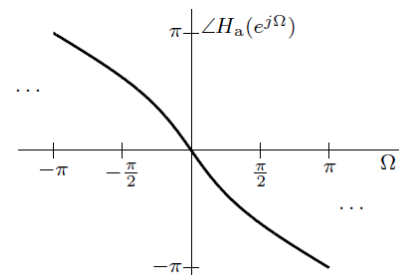
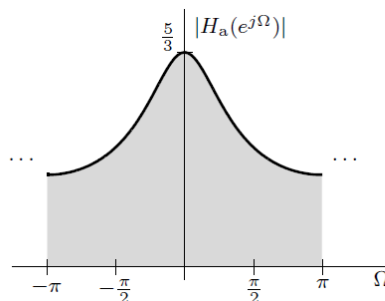
3)

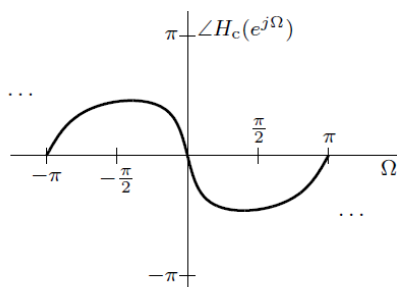
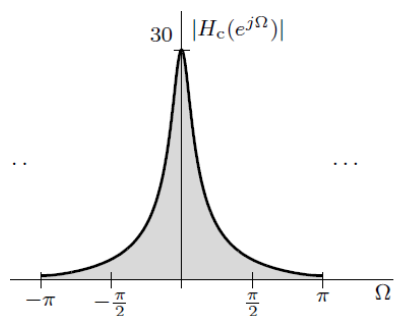
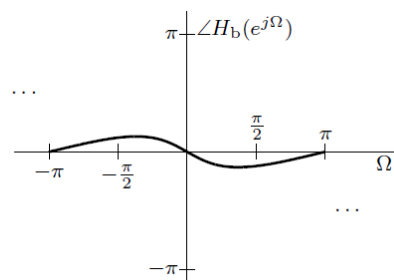
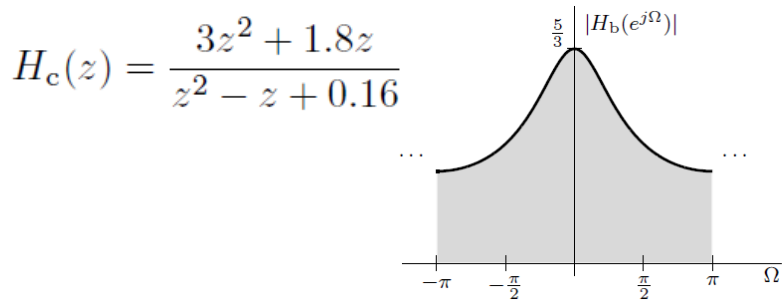


4)

$$H_a(z) = \frac{1}{z - 0.4}$$

$$H_b(z) = \frac{1}{1 - 0.4z^{-1}}$$

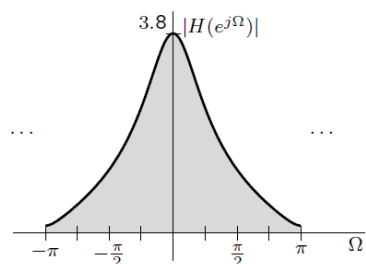




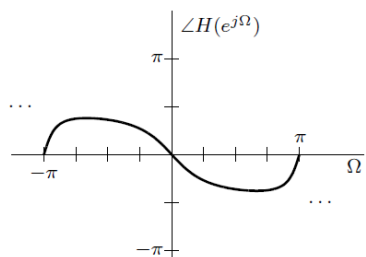
5) a) $H(z) = \frac{z + 0.9}{z - 0.5}$

Pole-zero plot for $H(z)$. The plot shows a pole at $z = 0.5$ (marked with a star) and a zero at $z = -0.9$ (marked with a dot). The x-axis is marked at -0.9 and 0.5 .

b) $H(\Omega) = \frac{e^{j\Omega} + 0.9}{e^{j\Omega} - 0.5}$



c) $y[n] = 3.8 + 1.203 \cos(0.5\pi n - 2.24)$



6) $\Omega_c = \frac{\pi}{4}$

$$H(z) = K \frac{(z - e^{j\pi/4})(z - e^{-j\pi/4})}{(z - |\gamma|e^{j\pi/4})(z - |\gamma|e^{-j\pi/4})}$$

